INCH-POUND

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MILITARY SPECIFICATION

ROLLER, VIBRATORY, SELF-PROPELLED

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a commercial, self-propelled vibratory roller.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-P-35	- Passivation Treatments for Corrosion-Resisting Steels.
TT-C-494	- Coating Compound, Bituminous, Solvent Type, Acid Resistant.
VV-F-800	- Fuel Oil, Diesel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895

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MILITARY

MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose.
MIL-R-3075 - Roller, Motorized, Road, Diesel- or-Gasoline Driven, Packaging of.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-A-46153 - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-A-52363 - Air Cleaner Intake: Dry Type.
MIL-C-53039 - Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-T-62314 - Test Equipment (Simplified) for Internal Combustion Engine (STE/ICE).
MIL-T-83133 - Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

MILITARY

MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-193 - Paint Procedures and Marking for Vehicles, Construction Equipment and Materials Handling Equipment.
MIL-STD-209 - Slinging and Tiedown Provision For Lifting and Tying Down Military Equipment.
MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-669 - Loading Environment and Related Requirements for Platform Rigged Airdrop Materiel.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-814 - Requirements for Tiedown, Suspension and Extraction Provisions on Military Material for Air Drop.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1366 - Material Transportation System Dimensional and Weight Constraints.
MIL-STD-1410 - Methods for Selection of Industrial Engines for End Item Application for Engine Classes and Type.
MIL-STD-1474 - Noise Limits for Army Materiel.
MIL-STD-1791 - Designing for Internal Aerial Delivery in Fixed-Wing Aircraft.
MS52149 - Battery, Storage, Lead Acid, Waterproof.
MS53052 - Bracket Assembly, Liquid Container, Five Gallon.

HANDBOOK

MILITARY

MIL-HDBK-157 - Transportability Criteria.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR (DOL)

29 CFR 1926.602 - Federal Motor Carrier Safety Regulation.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

US ARMY TANK-AUTOMOTIVE COMMAND (TACOM)

Report No. CR-82-588-003, STE/ICE Design Guide for Vehicle Diagnostic Connector Assembly.

DRAWINGS

TACOM

12258955 - Vehicle Test Card.

(Copies of the TACOM report and the drawing required by contractors in connection with specific acquisition functions should be obtained from the USA Tank Automotive Command, ATTN: AMSTA-MD, Warren, MI 48397-5000.)

ME

13229E8014 - Bracket Assembly Rifle Stowage.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FMT, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 698 - Moisture-Density Relations of Soils.
D 2487 - Classification of Soils for Engineering Purposes.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code - Steel.
D14.3 - Welding Earthmoving and Construction Equipment.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P. O. Box 351040, Miami, FL 33135.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

General Rules Governing the Loading of Commodities on Open-Top Cars and Trailers.

(Application for copies should be addressed to the Association of American Railroads, 50 F Street NW, Suite 7702, Washington, DC 20001.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J30 - Fuel and Oil Hoses.
J53 - Minimum Performance Criteria for Emergency Steering of Wheeled Earthmoving Construction Machines.
J154 - Operator Enclosures, Human Factor Design Considerations.
J164 - Radiators Caps and Filler Necks.
J209 - Instrument Face Design and Location for Construction and Industrial Equipment.
J386 - Seat Belts for Construction Machine.
J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.
J492 - Rivets and Riveting.
J514 - Hydraulic Tube Fittings.
J534 - Lubrication Fittings.
J536 - Hose Clamps
J541 - Voltage Drop for Starting Motor Circuits.
J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices.
J598 - Sealed Lighting Units for Construction and Industrial Machines.
J614 - Dipstick Marking.
J753 - Lubrication Chart - Construction and Industrial Machinery.
J833 - USA Male and Female Physical Dimensions for Construction and Industrial Equipment Design.
J898 - Control Locations for Off-Road Work Machines.

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J899 - Operator's Seat Dimensions for Off-Road, Self-Propelled Work Machines.
J917 - Marine Push-Pull Control Cables.
J925 - Minimum Service Access Dimensions for Off-Road Machines.
J1040 - Performance Criteria for Rollover Protective Structures (ROPS) for Construction, Earthmoving, Forestry, and Mining Machines.
J1127 - Battery Cable.
J1136 - Minimum Performance Criteria for Braking System for Self Propelled, Smooth Steel Wheel, Pneumatic Tired or Multipull Projection Steel Wheel Rollers and Compactors.
J1163 - Method for Determining Operator Seat Location on Agricultural and Construction Machines.
J1164 - Labeling of ROPS and FOPS.
J1362 - Symbols for Controls, Indicators, and Tell-Tales for Off-Road, Self-Propelled Work Machines.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC.

T&RA Year Book

(Application for copies should be addressed to Tire and Rim Association, Inc., Command Building, 34 North Hawkins Avenue, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Description. This specification covers a commercial, self-propelled, smooth drum, vibratory roller. The roller shall be diesel-engine driven and shall have a hydraulically operated hinge pin steering system. The roller shall have a strike-off blade and be equipped with a removable segmented pad foot shell kit. The roller shall be transportable by fixed wing aircraft and helicopter. In addition, the roller shall be capable of withstanding extraction parachute provisions. The roller shall have dual amplitude and dual frequency vibration operational characteristics. The roller shall be the current commercial production model of a standard product except for product improvements which have been incorporated on the commercial production units. The roller shall be equipped with all components furnished as standard equipment that are customarily offered with a roller of this type. For the purposes of this specification, the roller is defined as a single smooth drum with pad foot attachments.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.1.

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3.3 Material. Material not specified herein shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The roller shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the roller may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the roller may be newly fabricated from recovered materials to the maximum extent practicable, provided the roller produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the roller.

3.4 Maintainability. Provisions shall be made for ease in adjusting, servicing, or replacing parts and components. Access openings on the exterior of the roller shall have a removable or hinged cover.

3.5 Weight and dimensions. Roller characteristics shall conform to the following:

a. Operating weight with segmented pads, maximum - 16,000 pounds.
b. Smooth drum diameter, minimum - 46 inches.
c. Drum width, minimum - 66 inches.
d. Overall width, maximum - 96 inches.
e. Turning radius at drum edges, outside, maximum - 200 inches.
f. Gradeability, without vibration on hard surface, minimum - 20 percent.
g. Travel speed range, minimum - 0 to 6 miles per hour.
h. Curb clearance, minimum - 14 inches.
i. Oscillation, minimum - ±7 degrees.
j. Articulated steering, full power, minimum - left/right 28 degrees.

3.5.1 Height. The fully equipped roller less exhaust, stack, and rollover protection structures (ROPS) shall not exceed 109 inches to allow for air drop. The height shall be reducible to 90.5 inches or less. Reducible height shall be accomplished utilizing only common hand tools. Deflation of tires or removal of tires and rims is not permissible to reduce height.

3.6 Transportability. The roller shall be capable of being transported worldwide by highway, rail, marine, and air modes in C-130 and C-141 aircraft and to bee airmobile with CH-47 medium lift helicopter. The applicable

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transportability criterion is as set forth in MIL-STD-1791 and MIL-STD-1366. The roller shall withstand impact forces encountered in rail and air shipments without damage or permanent deformation. The roller shall not exceed the Gabarit International De Chargement (GIC) clearance dimensions shown in figure 1 for rail transport. The vehicle slinging and tiedown provisions shall withstand all forces to be encountered during loading or transportation (see 3.6.4 and 3.6.5). Preparation for internal air transport and reassembly after transport in C-130 and C-141 aircraft shall be accomplished in 60 minutes or less using only common hand tools (see 3.6.8).

3.6.1 Air transportability. The roller shall be transportable in the C-130, C-141 and C-5A aircraft. Transportability requirements shall be in accordance with MIL-STD-1791 and as specified herein. ROPS may be removed or relocated to meet the air transportability requirements using only a mechanic with common hand tools and a lifting device with operator. Removing or replacing the ROPS shall require not longer than 60 minutes. The ROPS shall have lifting eyes attached for ease of removal during air transport. The roller shall have 25 percent of its fuel supply and shall be fully serviced.

3.6.2 Rail impact. The roller shall be capable of withstanding shock loads resulting from rail impact testing in accordance with 4.4.2.2 without failure, damage, or permanent deformation.

3.6.3 Marine transportability. The roller shall be transportable by ships in accordance with MIL-HDBK-157.

3.6.4 Slinging provisions. The roller shall be provided with four sling provisions conforming to MIL-STD-209, type II equipment, class 1 or 3. The provisions shall enable the roller to be lifted in the normal traveling or operating position. Provisions shall be positioned so that the lifting strain shall be in line with the longitudinal axis of the eye of the provision. Provisions shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provision without weld failure or permanent deformation of the roller or the provision. Slinging provisions shall be labeled "LIFT HERE" in minimum 1/2-inch high letters.

3.6.5 Tiedown provisions. The roller shall be provided with integral provisions to permit tiedown of the roller to the floor or deck of the transportation medium. The tiedown provisions shall conform to MIL-STD-209, type II equipment, class 2 or 3 and shall withstand stresses in the amount and direction of pull specified without weld failure or permanent deformation of the roller. Tiedown provisions shall be labeled "TIEDOWN" in minimum 1/2-inch high letters.

3.6.6 Extraction parachute provision. A single fitting shall be provided on the roller for attachment of the parachute extraction system. The fitting shall conform to MIL-STD-814 and shall withstand stresses in the amount and direction of pull specified without weld failure or permanent deformation of the roller or the provision. The design ultimate strength of welded sections shall be four times the limit load. Low airdrop parachute extraction system (LAPES) fittings shall be provided unless vehicle design permits platforms extraction as specified in MIL-STD-814.

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3.6.7 Helicopter transportability. The roller shall be externally transportable by CH-47 helicopter in a single sortie. In the helicopter transportability configuration, the roller shall not exceed 16,000 pounds and must be fully operable. Disassembly of any components/parts is not allowed.

3.6.8 Airdrop. The roller shall conform to MIL-STD-669 for airdrop from C-130 and C-141 aircraft without breakage or permanent deformation of any structural member or component, or loss of the roller to perform all functions specified herein. When rigged for airdrop in accordance with MIL-STD-669, the roller shall meet the established tipoff curves for the C-130 and C-141 aircraft as defined in MIL-STD-1791. Disassembly and reassembly for low velocity airdrop and LAPES shall be accomplished in a combined time of not more than 60 minutes by no more than 2 people. Tires or wheels shall not be removed for airdrop. The suspension system and tires shall be capable of withstanding four-inch vertical travel upon impact when sitting on six inches of honeycomb material.

3.7 Performance. The roller shall:

a. Be capable of compacting soil conforming to type GC/SC (Unified Soil Classification System) to at least 95 percent Proctor (ASTM D 698) in compacted lifts of not less than 12 inches.
b. During smooth drum operation the roller shall be capable of compacting 500 cubic yards of granular type soils (type GC) to at least 95 percent Proctor in compacted lifts of not less than 12 inches in one hour using the hydraulic strike-off blade to control lift thickness.
c. During pad foot operation the roller shall be capable of compacting 350 cubic yards of cohesive soils (type SC) to at least 95 percent Proctor (ASTM D 698) in compacted lifts of not less than 12 inches in one hour using the hydraulic strike-off blade to control lift thickness.
d. Be capable of negotiating a 20 percent grade over firm soil.
e. Be capable of operating in ambient temperatures of not less than 120 °F without exceeding the engine manufacturer's recommended limits for radiator top tank temperature, engine oil temperatures (sump and gallery), and component manufacturer's hydraulic fluid temperature limits.
f. Be capable of two frequencies and amplitude. Frequencies shall not be adjustable by changing the roller's engine speed. High amplitude centrifugal force shall be at a minimum, 26,000 pounds.

3.8 Human factors. The characteristics of the roller shall provide for operation by personnel ranging from the 5th percentile female through 95th percentile man in accordance with SAE J833. Minimum service access dimensions shall be in accordance with SAE J925 for arctic clothed personnel. In addition, the roller shall be in accordance with the following SAE specifications J154, J209, J898, J899, and J1362.

3.9 Safety. All reciprocating, rotating, or moving parts of all components shall be guarded when such parts may be inadvertently contacted by operating and maintenance personnel or otherwise create a hazard. All hot surfaces, including exhaust pipes and other lines which may be subject to high temperatures and exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto engine, exhaust, or electrical equipment. Exhaust or discharges from equipment shall be directed so that they do not

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endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided.

3.9.1 Ladders, steps, and handholds. Ladders, steps, and handholds shall be installed to assist the entrance of and exit from the operator's compartment.

3.10 Noise limits. Noise levels produced by the roller shall comply with the requirements of MIL-STD-1474, except as indicated herein.

a. The noise level at the operator's position and at one meter from the perimeter of the roller at all other locations shall not exceed Category D of table I (85 dB(A). MIL-STD-1474, 5.1.1.2, 5.2, 5.3 and 5.4 shall not apply. Compliance and non-compliance with the required noise limits shall be documented based upon test requirements of 4.4.4 and the data requirements of MIL-STD-1474, 5.5.
b. The following table I shall be used in lieu of table II contained in MIL-STD-1474B:

TABLE I. Steady-state noise limits (dB) for personnel-occupied areas.

	Limit category					
	A*	B*	C*	D*	E**	F**
A-weighted limit ([dB(A)]	108	100	90	85	75	65

3.11 Body and frame. The body and framing shall be two separate bodies connected at a center hinge point to provide articulated center hinge point steering.

3.12 Engine. The roller engine conforming to MIL-STD-1410 shall be a commercial model, diesel-type, having horsepower, torque, and speed characteristics meeting all performance requirements specified herein. The engine shall operate on JP-8 turbine fuel or diesel fuel conforming to MIL-T-83133 or VV-F-800, respectfully.

3.12.1 Power and speed rating. The engine power and speed ratings shall be such that operation of the roller under any conditions specified herein will not require horsepower in excess of the engine's intermittent horsepower rating for the applicable governed speed, as established by the engine manufacturer.

3.12.2 Governor. A governor shall be provided to limit the engine speed to that recommended by the engine manufacturer when the roller is operated under any condition specified herein.

3.12.3 Radiator. The radiator assembly, when required, shall be heavy-duty, when used as a part of the cooling system. Where the top of the radiator is lower than the top outlet of the engine, a surge tank shall be used. The radiator cap shall meet the requirements of SAE J164 and shall be accessible

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without removing the engine hood. A warning sign "CAUTION - DO NOT OPEN WHEN HOT" shall be placed near the radiator cap. The radiator shall be adequately protected.

3.12.3.1 Gaskets, seals, and packing. Gaskets, seals, and packing used in the coolant system shall be made of materials that resist deterioration from mixtures of water and ethylene-glycol-based antifreeze conforming to MIL-A-46153.

3.12.3.2 Engine cooling. The cooling system shall maintain coolant at a temperature below the boiling point when working at rated rpm and operating at 10,000 feet above sea level or ambient temperature up to 125 °F.

3.12.3.3 Engine coolant. Antifreeze shall conform to MIL-A-46153. The mixture of coolant shall protect the engine cooling system to -50 °F.

3.12.3.4 Coolant hose clamps. All hose clamps shall be in accordance with SAE J536, type F.

3.12.5 Exhaust system and exhaust cover. The engine shall have the manufacturer's current standard exhaust system.

3.12.6 Air cleaner. The engine shall have a heavy-duty, dry-type air cleaner conforming to MIL-A-52363, class II or III. The air cleaner shall be a dual element. When the engine has a turbocharger, the cleaner shall include a dust scavenger. The air cleaner capacity shall be in accordance with the engine manufacturer's recommendations. The air cleaner shall be readily accessible without the need to move or reach around other parts. An air cleaner intake-restriction indicator shall be provided and shall be visible from the operator's compartment. The intake-restriction indicator shall have graduated markings with continuous reading on remaining air cleaner service with and without the engine running. The indicator also shall have reset capabilities.

3.12.6 Oil filter. Heavy-duty, full-flow oil filters shall be furnished in accordance with the engine manufacturer's recommendations. A label specifying filter type (see 3.26.2) shall be placed near the point of filter installation. When applicable, oil lines to and from the filters shall be installed to minimize vibration. These lines shall be of flexible hose meeting the requirements of SAE J30. The oil filter shall be readily accessible without the need to move or reach around other parts.

3.12.7 Engine access. Engine hood or side panels shall be furnished for engine access.

3.12.8 Dipstick marking. The engine oil dipstick shall be marked in accordance with SAE J614.

3.13 Operator seat. A cushioned operator seat shall be furnished. The seat shall be located in accordance with SAE J1163 to permit access by 95 percent of the US male and female operator population in accordance with 3.8. An adjustable, weather-resistant seat belt with buckle, conforming to SAE J386, type I, shall be furnished. The operator's seat shall be as specified in 3.8.

3.14 Controls. Each control shall be the manufacturers standard and shall have labels embossed or depressed on or near the control. Labeling shall be in

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accordance with the criteria of 3.8. Labels (see 3.26.2) shall be visible to the operator from a standard operating position. All roller controls shall be located in the operator's compartment. Electric controls shall be weathertight. All hand-operated controls shall be operable by persons wearing cold weather or wet gloves. Mechanical push-pull remote controls, when furnished, shall be waterproof, heat-resistant, and antifriction conforming to requirements for marine cables specified in SAE J917, except as specified herein. As a minimum, all exposed parts of the control cables shall be made of corrosion-resistant steel (except 400 series stainless steel) that has been passivated in accordance with QQ-P-35. The cable ends shall be covered with a polyethylene or vinyl jacket. The cable ends shall be able to actuate all levers through their respective arcs without binding. Bend radii shall be as recommended by the cable manufacturer. Drain holes with one-way membranes shall be made for all weather-tight electric controls or compartments.

3.14.1 Engine run control. An engine run control shall be furnished and shall be the keyless type. It shall have not less than two positions: "on" and "off". The switch shall control all electrical monitors and cab electrical equipment, except the horn and lights.

3.14.2 Instrumentation. The roller shall have lights, gages, and instruments for monitoring all roller systems. All gages shall be weathertight and shall have provisions for adequate lighting for night operation. Instrumentation shall be in accordance with SAE J209 as specified in 3.8.

3.15 Rollover protection structures (ROPS). The roller shall have ROPS with overhead protection as specified in SAE J1040. The ROPS shall be removable using common hand tools and a lifting device. The weight of the structure shall be labeled in accordance with SAE J1164 to meet transportability requirements (see 3.6.1), if required.

3.16 Propulsion system. The traction drive unit shall be equipped with a hydrostatic drive and, if normally supplied, a manual transmission. The roller shall be equipped with a single forward and reverse speed control which shall provide infinite speed selection. A separate lever shall be supplied for gear selection when a manual transmission is furnished. The propulsion system shall provide for dynamic braking. In addition to drum drive, wheel drive shall be provided. Drive axle for rear wheels shall be equipped with differential lock or limited slip.

3.17 Vibration system. The vibration system shall be of the direct hydraulic drive type. No external exciters are permitted. The vibrating mechanism shall be contained within the drum.

3.18 Steering system. The steering system shall provide for full power hydraulic steering of the articulated type, with horizontal oscillation of the drum provided at the steer pivot point. The steering system shall have an on-board steering locking device to prevent articulation during transport. In the event of power assist failure, the steering shall comply with the emergency steering requirements of SAE J53.

3.19 Braking system. The roller shall have a braking system as specified in SAE J1136 and shall meet or exceed all test requirements therein. The roller shall have warning lights and an audible alarm when the emergency stopping system

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is activated. In the event of power assist failure, the brakes shall be capable of safely stopping the roller with a pedal force of not more than 70 pounds. A parking brake and emergency stopping brake shall be provided.

3.20 Tires and rims. Tires and rim assemblies shall be in accordance with the Tire and Rim Association Year Book. The tires on the rear section of the roller shall be of the tubeless, earthmoving, traction tread type. The tires shall have a size and a load carrying capacity, in accordance with the Tire and Rim Year Book, equal to or greater than the maximum roller loading imposed on the tires. The Tire and Rim Association Yearbook recommended tire pressure shall be stenciled as "TP (specify) LBS" in 1-inch high letters above or adjacent to each tire. Rubber based components shall be not more than four calendar quarters in age from date of Vulcanization to date of acceptance by the Government.

3.21 Strike off blade. The roller shall be equipped with a hydraulically controlled strike off blade which shall be part of the roller specified herein. This blade shall be equal to or greater than the overall width of basic machine. A "float" position shall be supplied with detente on the control to maintain desired operation. Adjustable skids shall be attached to the blade to enable operator to set the blade at a predetermined strike off height to maintain a uniform lift thickness.

3.22 Compaction monitoring device. The roller shall have a comparative compaction monitoring device capable of detecting soil density during normal compaction operations. The device shall monitor relative load-bearing conditions for various soil types during compaction. At the operator's station, an analog or readout provisions shall continually monitor the soil and the compaction effort. The readout shall show when maximum compaction is attained. The device shall receive power from the compactor's electrical system. The sensors and transducers shall be protected during operations and cleaning. All exposed wiring shall be protected in conduit.

3.23 Pad foot shell kit. The roller shall be supplied with a pad foot shell kit. The shell kit shall be designed for installation in segments. The segments shall have provisions for lifting. The roller shall be equipped with lifting chains or cables for segment handling. The chains or cables shall be stowed in the roller's tool and accessory box. The shell kit shall be capable of being installed or removed by two men in two hours using common hand tools and material handling equipment. The weight of each segment and installation procedures shall be embossed on labels (see 3.26.2). The labels shall be installed near the front of the vehicle within eye level view when the operators are standing on the ground.

3.24 Lubrication. All surfaces requiring lubrication shall be lubricated. A lubrication chart identical to the chart provided in the operator and maintenance manuals and in accordance with SAE J753 shall be furnished. The chart shall identify applicable military lubricants and appropriate ambient temperature ranges and shall be mounted as specified in 3.26.2.

3.24.1 Lubricants. All parts and components shall be designed so that they are properly lubricated with specified military lubricants. The engine shall operate on oil conforming to MIL-L-2104. Initial lubrication and all subsequent lubrication shall be with military lubricants conforming to MIL-L-2104, MIL-L-2105, and MIL-G-10924. All components shall be filled to the operating

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level with appropriate operational grades of military lubricants designated for use in the temperature ranges of the destination of the delivered system. Labels shall be as specified in 3.26.2.

3.24.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected, accessible position. Fittings shall be accessible to a grease gun with a 10-inch flexible extension. Fittings shall be accessible without removing or adjusting accessories or parts. Fittings requiring the operator to crawl into areas to lubricate components shall have a metal or flexible line extended to an exterior location for ease of lubrication. Extended lubrication points shall be accessible to an operator standing on the ground.

3.24.3 Fluid drains. The roller shall have provisions to drain the required service fluids without trapping the fluids in the supporting structures. If extensions are required, they shall be labeled as specified in 3.26.2 as to the type of fluid being drained. All fluid drains shall be capable of completely draining the fluid into a suitable container.

3.24.4 Pressure-release device. An automatic pressure-release device shall be provided where pressure-lubricating equipment could damage grease seals or other parts.

3.24.5 Oil sampling valve. An oil sampling valve shall be provided on the engine. The valve shall be manually operated and shall automatically close after release. The valve shall be made of material resistant to corrosion such that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running. The "open" position of the valve control shall be labeled "OPEN".

3.25 Electrical system. The manufacturer's standard or optional electrical system shall be provided that meets the requirements specified herein. A 24-volt starting system shall be furnished.

3.25.1 Alternator. The manufacturer's high-output option alternator shall be furnished with not less than 50 amps.

3.25.2 Starting system. The starting system shall be a 24-volt system in accordance with the engine manufacturer's recommendations.

3.25.3 Batteries. Batteries furnished shall be the contractor's standard or optional batteries for cold weather starting. If batteries are placed in a battery box, there shall be a warning placed on the box to warn of the presence of toxic fumes and the explosive hazards.

3.25.4 Battery mounting. Batteries shall be located so they can be cleaned, serviced, and removed from ground level without removing any component except the quick-release battery box cover. When a battery box is furnished, battery supports, hold-downs, and area around the battery compartment which could be damaged by dripping or seeping acid shall be protected with a coating conforming

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to TT-C-494, type II. The battery shall be mounted so it does not interfere with access to engine components and accessories. Battery mountings shall completely support the battery base. Battery restraining clamps shall be provided to hold the battery base and to hold the battery in a fixed position. The battery compartment (if furnished) shall have provisions for drainage and for gas venting at or near the top of the compartment. The cover and cable position shall be protected by rubber grommets or by insulated passages at the entry to the battery box and frame. Drain holes shall be provided in the battery compartment or mount.

3.25.5 Battery cable. Battery cables meeting the requirement of SAE J1127 shall be furnished. The voltage at the storage battery terminals and at the starting motor, including connections, shall not differ more than those shown in table I of SAE J541. Positive and negative cable terminals shall be identified with a red sleeve labeled "+" on the positive cable and a black sleeve labeled "-" on the negative cable. The positive cable shall have a captive rubber cap for placement over the positive battery terminal connection.

3.25.6 Circuit protection. Each electrical circuit shall have a protection device. The device may be circuit breakers or replaceable fuses. The device shall be accessible from operators platform without removing any other component. The protection devices shall have labels (see 3.26.2) describing the function of the device.

3.25.7 Instrument lighting. The roller shall have gage lighting or indicators which are visible in all ambient lighting conditions.

3.25.8 Horn. The roller shall have a horn. The horn shall be a minimum of 10 dB(A) louder than the engine at full-governed speed. A reverse signal alarm, audible type conforming to 29 CFR 1926.602 shall be supplied.

3.25.9 Electromagnetic interference. The electromagnetic interference emission and susceptibility characteristics of the roller shall conform to SAE J551, devices category.

3.25.10 Lamps. The roller shall have two floodlamps with protective guards mounted on the front and two floodlamps mounted on the rear in accordance with SAE J598. The lamps shall be brush-guard protected. The sealed units shall be replaceable without special tools.

3.25.11 Hourmeter. An electrical hourmeter registering the number of engine operating hours, up to a minimum of 9999 hours, shall be provided at the operator's station. The hourmeter shall be accessible without removing any roller component.

3.25.12 Diagnostic connector assembly (DCA). The roller shall incorporate an easily accessible DCA in the operator's cab for interface with the Government's Simplified Test Equipment/Internal Combustion Engine (STE/ICE) in accordance with Report No. CR-82-588-003 and MIL-T-62314. As a minimum, the DCA shall be capable of monitoring the functions defined in table II of Report No. CR-82-588-003 and alternator/generator output voltage, alternator/generator negative cable voltage drop, fuel solenoid voltage, starter voltage, starter current first peak and average, starter circuit resistance, battery internal resistance change, fuel supply pressure, and fuel filter restriction (Delta P). Test points identified

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to monitor functions in the TK mode shall be accessible without the removal of major assemblies and installed attachments. Fittings/adapters required for the interface with the STE/ICE test set when monitoring in the TK mode shall be permanently installed on the roller. Adapters in the STE/ICE test set shall readily mate with test point connections in the roller. A separate wiring harness shall be provided for the DCA and shall include all wiring and necessary hardware to perform required capabilities. The contractor shall also provide STE/ICE vehicle test cards in the format identified in Drawing No. 12258955 addressing both DCA and TK measurements.

3.26 Plates. The following plates shall be installed on the end item.

3.26.1 Identification marking. Each roller and attachment removable for either surface or air transportability (e.g., cab, ROPS), shall be identified in accordance with MIL-STD-130 on plates conforming to MIL-P-514. The plates shall be attached by screws, drive screws, bolts, or rivets in conspicuous, protected locations and shall indicate the silhouette of the roller in the transport position showing the center of gravity. The silhouette shall show the side and end views.

3.26.2 Instruction plates/labels. The roller shall have instruction plates and labels conforming to MIL-P-514. The plates and labels shall include warnings, cautions, and diagrams describing any special or important procedures to be followed in assembling, operating, lubricating, and servicing the roller.

3.27 Bracket assembly, rifle stowage. A rifle stowage bracket assembly conforming to drawing 13229E8014 shall be furnished and mounted by the contractor. The location of the bracket shall be that it is accessible to the operator and provides clearance for the rifle to be removed. The bracket shall be located away from operators controls and shall not impair the vision of the operator.

3.28 Towing lugs. The roller shall have towing provisions. Each set of towing lugs, i.e., front and rear, shall support one and a half times the roller weight. The roller shall have two lugs on the front and two lugs on the rear that will not create a safety hazard for operator and maintenance personnel.

3.29 Toolbox. A toolbox to hold all tools shall be provided. The toolbox shall have drain holes to prevent moisture from accumulating.

3.30 Drain holes. Drain holes shall be provided where applicable to prevent collection or entrapment of water or other fluids.

3.31 Vehicle weight classification sign kit. The contractor shall apply the vehicle weight classification numbers to each vehicle using a vehicle weight classification number kit furnished by the Government (see 3.33). The kit shall be located on the front of the vehicle in a location approved by the contracting officer. The contracting officer shall assign the classification numbers to be displayed on the vehicle. Instructions for changing the numbers, based on the vehicle configuration, shall be provided.

3.32 Case. The contractor shall furnish the fasteners and shall install the Government-furnished case (see 3.33) in the operator's compartment.

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3.33 Government-furnished property. When specified (see 6.2), the following property in the quantity indicated will be furnished by the Government (see 6.4).

Item No.	Description	Quantity
1	Camouflage paint pattern.	As required
2	Cases complete with maintenance forms Case NSN 7520-00-559-9618.	As required
3	Vehicle Weight Classification Sign Kit NSN 9905-00-565-6267.	As required
4	Report CR 82-588-003 STE/ICE Design Guide for Vehicle Diagnostic Connector.	As required

3.34 Decontaminating apparatus, portable (DAP), M13. The roller DAP mounting bracket (MS53052-1) shall be installed in a manner which shall not interfere with the normal operation of the equipment. The DAP shall be mounted such that the user will not be required to lift the device more than 3 feet above any standing surface. An upright installation is preferred. Horizontal mounting shall be used only with the approval of the contracting officer. The DAP device has a flash point of 165 °F and shall be installed away from exhaust or heat sources.

3.35 Treatment and painting. All interior and exterior portions of the roller normally painted and all exposed surfaces that require protection from the environment shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G. Unless otherwise specified (see 6.2), the top coat color shall be camouflage green 383 conforming to MIL-C-46168 or MIL-C-53039. When specified (see 6.2), the roller shall be overcoated in accordance with the Government furnished camouflage pattern with MIL-C-46168 or MIL-C-53039 top coat of the colors specified in the camouflage patterns except the first article may be camouflage green 383. All surfaces of the roller which exceed 400 °F during operation shall not be primed or painted as specified above but shall be painted in the appropriate color of the base coat or camouflage pattern scheme, if applicable, with heat resisting paint identified in MIL-STD-193. All hardware not normally painted shall be treated to provide limited reflectivity. The entire roller including under the hood, except for the engine and transmission as noted below, shall be painted with paint conforming to MIL-C-46168 or MIL-C-53039. The engine and transmission, if unexposed, shall be painted or provided a non-reflective surface; if exposed, the engine and transmission shall be painted in accordance with MIL-C-46168 or MIL-C-53039. There shall be no exposed unpainted ornamental trim.

3.35.1 Marking. Marking and registration shall be in accordance with MIL-STD-642. Color shall be lusterless black.

3.35.2 Camouflage pattern data. When specified (see 6.2), the contractor shall provide camouflage pattern data.

3.36 Workmanship.

3.36.1 Casting and forgings. All parts, components, and assemblies of the roller, which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

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3.36.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subject to proof and service loading. Parent material, weld-fill metals, and fabrication techniques shall be required to enable the roller to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be as close as possible, and in no event shall they be separated by more than 3/16-inch, unless appropriate bridging techniques are used. The welding process used in fabricating the roller shall be the contractor's option. Welding procedures shall be qualified in accordance with applicable ASME or AWS code, and all welding and welds shall be in accordance with ASME or AWS code.

3.36.2.1 Welders. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall ensure that the welder or welding operator has passed qualification tests, as prescribed by one of the following listed codes for the type of welding operations to be performed and that such qualification is effective.

AWS D1.1 Structural Welding Code - Steel.
AWS D14.3 Welding Earthmoving and Construction Equipment.
ASME Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers and Welding and Brazing Operators.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to Government approval, the contractor's standard welder qualification may be substituted in lieu of the above codes. In the event of evidence of poor welds, the Government reserves the right to require recertification of any welder or welding operator. The test results shall be made available for review by the contracting officer or his representative.

3.36.2.2 Weld quality. Weld quality shall be as follows:

a. The contractor shall be responsible for the roller conforming to the inspection requirements specified in section 4.
b. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.
c. Undercut in weldments shall be minimized and shall not be more than 1/32-inch deep.

3.36.2.3 Welding practices. Weld practice shall be as follows:

a. Preheating of materials being welded and maximum interpass temperature during welding shall be in accordance with the contractor's standard practice.
b. Tack welds shall be subject to the same quality requirements as final welds.
c. Work shall be positioned for flat welding, whenever practicable.
d. Procedures and sequences shall minimize distortion and shrinkage.

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3.37 Bolted connections. Bolt holes shall be formed accurately and shall have the burrs removed. Washers, lockwashers, or other positive locking devices shall be provided, where necessary. Matching thread areas securing bolts or capscrews shall conform to SAE J429 and shall withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.38 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neat, concentric with the rivet holes, and in full contact the surface of the member, and shall be in accordance with SAE J492.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article roller shall be inspected as follows:

a. Examination (see 4.3).
b. All tests under 4.4.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to each item being offered for acceptance under the contract. Failure of the roller to pass the examination tests or inspection shall be cause for rejection. Quality conformance shall consist of the following:

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a. Examination (see 4.3).
b. Tests (see 4.4.1.1).
c. Inspection of preservation, packaging, packing, and marking for shipment and storage (see 4.5).

4.2.3 Test rejection criteria. Throughout all tests specified herein the roller shall be observed for the following conditions, any of which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein.
b. Spillage or leakage of any fluid, including fuel, coolant, lubrication, or hydraulic fluid under any operational condition.
c. Structural failure, evidence of impending failure, or permanent deformation of component.
d. Misalignment of any component.
e. Conditions presenting a safety hazard to personnel during operation, service, or maintenance.
f. Overheating of the engine or any other roller components.
g. Evidence of excessive wear.
h. Interference between roller components or between the roller and the ground, with the exception of normal contact by the rollers.
i. Evidence of instability in handling during cornering, braking, and while traversing rough terrain.

4.3 Examination. Each roller shall be examined for compliance with the requirements specified in section 3 of this specification. The first article and quality conformance article shall be examined in accordance with table II. An "X" in the column indicates the examinations that shall be conducted. Noncompliance with any specified requirement shall be cause for rejection.

TABLE II. Examination.

First Article	Qual. Conf.	Description	Requirement Paragraph
X	X	101. Materials not as specified	3.3
X	X	102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments.	3.3.1
X		103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.2
X	X	104. Contractor does not have documentation available for identification of materials, material finishes, or treatment.	3.3.3
X	X	105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the roller.	3.3.4
X		106. Maintainability not as specified.	3.4
X	X	107. Weight and dimensions not as specified.	3.5
X		108. Height not as specified.	3.5.1
X		109. Transportability not as specified.	3.6

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TABLE II. Examination. (continued)

First Article	Qual. Conf.	Description	Requirement Paragraph
X		110. Air transportability not as specified.	3.6.1
X		111. Marine transportability not as specified.	3.6.3
X		112. Helicopter transportability not as specified.	3.6.7
X		113. Airdrop requirements not as specified.	3.6.8
X		114. Human factors not as specified.	3.8
X	X	115. Safety not as specified.	3.9
X		116. Ladders, steps, and handholds not as specified.	3.9.1
X		117. Body and frame not as specified.	3.11
X		118. Power and speed rating not as specified.	3.12.1
X		119. Governor not as specified.	3.12.2
X		120. Radiator not as specified	3.12.3
X		121. Gaskets, seals, and packing not as specified.	3.12.3.1
X	X	122. Engine coolant not as specified.	3.12.3.3
X		123. Coolant hose clamps not as specified.	3.12.3.4
X		124. Exhaust system and exhaust cover not as specified.	3.12.4
X	X	125. Air cleaner not as specified.	3.12.5
X		126. Oil filter not as specified.	3.12.6
X		127. Engine access not as specified.	3.12.7
X		128. Dipstick marking not as specified.	3.12.8
X		129. Operator seat not as specified.	3.13
X		130. Controls not as specified.	3.14
X		131. Engine run control not as specified.	3.14.1
X		132. Instrumentation not as specified.	3.14.2
X		133. Rollover protective structures not as specified.	3.15
X		134. Propulsion system not as specified.	3.16
X		135. Vibration system not as specified.	3.17
X		136. Steering system not as specified.	3.18
X		137. Braking system not as specified.	3.19
X	X	138. Tires and rims not as specified.	3.20
X		139. Strike off blade not as specified.	3.21
X		140. Compaction monitoring device not as specified.	3.22
X		141. Pad foot shell kit not as specified.	3.23
X	X	142. Lubrication not as specified.	3.24
X	X	143. Lubricants not as specified.	3.24.1
X	X	144. Lubrication fittings not as specified.	3.24.2
X	X	145. Fluid drains not as specified.	3.24.3
X		146. Pressure-release device not as specified.	3.24.4
X	X	147. Oil sampling valve not as specified.	3.24.5
X		148. Electrical system not as specified.	3.25
X		149. Alternator not as specified.	3.25.1
X		150. Starting system not as specified.	3.25.2
X		151. Batteries not as specified.	3.25.3
X		152. Battery mounting not as specified.	3.25.4
X		153. Battery cable not as specified.	3.25.5
X		154. Circuit protection not as specified.	3.25.6

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TABLE II. Examination. (continued)

First Article	Qual. Conf.	Description	Requirement Paragraph
X		155. Instrument lighting not as specified.	3.25.7
X		156. Horn not as specified.	3.25.8
X		157. Lamps not as specified.	3.25.10
X		158. Hourmeter not as specified.	3.25.11
X	X	159. Diagnostic connector assembly not as specified.	3.25.12
X	X	160. Identification marking not as specified.	3.26.1
X	X	161. Instruction plates/labels not as specified.	3.26.2
X	X	162. Rifle stowage bracket assembly not as specified.	3.27
X		163. Towing lugs not as specified.	3.28
X	X	164. Toolbox not as specified.	3.29
X		165. Drain holes not as specified.	3.30
X	X	166. Vehicle weight classification sign not applied as specified.	3.31
X	X	167. Maintenance form case not installed as specified.	3.32
X	X	168. Decontaminating mounting bracket not as specified or not installed as specified.	3.34
X	X	169. Treatment and painting not as specified.	3.35
X	X	170. Color not as specified.	3.35
X	X	171. Camouflage pattern not as specified (when required).	3.35
X	X	172. Marking and registration not as specified.	3.35.1
X		173. Casting and forgings not as specified.	3.36.1
X		174. Welding not as specified.	3.36.2
X		175. Welders or welding operators not qualified as specified.	3.36.2.1
X	X	176. Weld quality not as specified.	3.36.2.2
X		177. Welding practices not as specified.	3.36.2.3
X		178. Bolted connections not as specified.	3.37
X		179. Riveted connections not as specified.	3.38

4.4 Tests.

4.4.1 First article test. All first article tests shall bed run using JP-8 fuel conforming to MIL-T-83133.

4.4.1.1 Run-in test. Rollers produced in fulfillment of the contract or order shall be completely assembled, adjusted, lubricated, and otherwise serviced for operation. A run-in test shall be conducted in accordance with the manufacturer's recommendations. During this test, all controls, adjusting mechanisms, brakes, hydraulic systems, electrical accessories, instruments, and other accessories shall be operated to ensure against restrictions, malfunctions, or leaks. Run-in tests shall be run using either JP-8 or diesel fuel conforming to MIL-T-83133 and VV-F-800, respectively.

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4.4.2 Transportability. The roller shall be tested for transportability as follows:

4.4.2.1 Air transportability. The roller shall be measured to prove conformance to 3.6.1. Disassembly of the ROPS shall be in accordance with the contractor's instructions. Nonconformance to the requirements of 3.6.1 shall constitute failure of this test.

4.4.2.2 Rail impact test. The roller shall be tested in accordance with MIL-STD-810 to prove conformance to 3.6.2. The rail cars shall be equipped with standard friction draft gears as defined by the Association of American Railroads (AAR) rules (without coupler cushioning devices) and conventional underframes. Inability to meet the requirements of 3.6.2, any permanent deformation or failure or inoperability after test shall constitute failure of this test.

4.4.2.3 Slinging and tiedown provisions. The roller shall be tested for compliance with 3.6.4 and 3.6.5. Failure to meet the requirements specified shall be cause for rejection of this test.

4.4.2.4 Extraction parachute provision. The extraction parachute provision shall be demonstrated to prove conformance to 3.6.6. Apply a force equal to 1.5 times the limit load specified in 3.6.6. Evidence of weld failure or permanent deformation of the extraction parachute provision shall constitute failure of this test.

4.4.2.5 Airdrop. The roller shall be subjected to simulated airdrop impact test in accordance with MIL-STD-669 to prove conformance to 3.6.8. This test shall be conducted by the Government at a Government facility, however, acceptance of the first article models shall be dependent of the successful completion of this test. The test shall consist of the following.

a. Rig the roller in accordance with MIL-STD-669 and confirm that the rigged weight and dimensions conform to the requirements of MIL-STD-1791.
b. Drop test the roller in accordance with MIL-STD-669.
c. After the first drop, the roller shall be de-rigged, examined, started and driven forward and backward a few feet, and all controls shall be actuated.
d. Re-rig and drop test the roller a second time in accordance with MIL-STD-669.
e. After de-rigging, the roller shall be demonstrated for 8 hours performing its intended function(s).

Failure to meet the weight and dimensional requirements of MIL-STD-1791, weld failure, leakage, permanent deformation of any structural member or component, inability to reassembly to components removed for airdrop, or inability of the roller to perform its intended functions shall constitute failure of this test. Failure to complete disassembly and reassembly in 60 minutes combine time by not more than 2 people constitutes failure of this test (see 3.6).

4.4.3 Performance. The fill material shall be as specified in 3.7 and group GC/SC of the Unified Soil Classification System in accordance with ASTM D 2487. The compacted fill strip shall be rectangular and not less than 31 feet wide and 1000 feet long. The fill shall be placed prior to the start of compaction

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operation. Before testing is started, moisture content shall be determined and a test strip of the material shall be compacted using the vibratory roller to determine proper drum configuration (pad foot or smooth), vibration, amplitude, speed and uncompacted lift thickness. Hauling equipment shall not be run over spread, uncompacted fill material before or after dumping the material. Spreading of fill material shall be accomplished with the strike off blade mounted on the compactor prior to compaction test. One hour shall be allotted to spreading material prior to compaction test. Water distribution equipment may be used as required. The material to be compacted shall be of sufficient thickness to give a compacted lift of not less than 12 inches. The laboratory density of the soil to be compacted shall be determined in accordance with ASTM D 698. The Government reserves the right to witness any laboratory testing. When the contractor feels that 350 compacted cubic yards of fill material has been compacted with pad foot compactor to a density of 95 percent of standard Proctor values, they may stop compaction operations. Time used in spreading or density testing shall not be counted as compaction time. Two density samples shall be taken at each 100 foot station. The first sample shall be taken 10 feet left of the centerline and the second sample 12 feet right of the centerline. This pattern shall be repeated for the length of the fill. In addition, other samples shall be taken at random locations as deemed necessary by the Government. The Government reserves the right to modify sampling pattern as necessary. Volume of compacted soil shall be determined as follows:

a. Stake out length of fill in 100 foot stations.
b. Using surveyor's level and rod, determine the elevations of left edge, right edge, and centerline at each station prior to placing the fill material and after compaction.
c. Determine the compacted width at each station.
d. Calculate the volume using the average and area method.

Density testing shall be accomplished by a nuclear density meter. The nuclear density tester shall have been calibrated against the soil to be compacted, if applicable. The Government shall witness such calibration. An average density of less than 95 percent standard Proctor, less than 350 compacted cubic yards at 12 inch lift thickness in one hour, of operation other than specified in 3.7 shall constitute failure of this test.

4.4.3.1 Smooth drum. The roller shall be tested to prove capability of compacting 500 cubic yards of granular type soils (type GC) to at least 90 percent Proctor in compacted lifts of not less than 12 inches in one hour using the hydraulic strike-off blade to control lift thickness. Nonconformance to 3.7 shall constitute failure of this test.

4.4.4 Sound level. Noise levels shall be measured in accordance with MIL-STD-1474, 5.1.2 through 5.1.5 and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.4.5 Compaction monitoring device. The roller shall be operated over an area of fill selected by the contractor until the device indicates that maximum compaction has been reached. The compacted area shall be tested using a nuclear density device to determine maximum density. The roller shall then be operated until the monitoring device indicates compaction has decreased. The nuclear

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density device shall be used to determine actual density. If the monitoring device indicates a lower reading and the density of the fill material has increased, it shall constitute failure of this test.

4.4.6 Electromagnetic interference. The roller shall be tested in accordance with SAE J551. Nonconformance to 3.25.9 shall constitute failure of this test.

4.4.7 DCA capability. The contractor shall perform a 100 percent visual inspection of all DCA wiring harnesses prior to installation on the roller to ensure wire and connections are free of error. After installation on the roller, test the DCA harness (see Report No. CR-82-588-003) on each vehicle with a DCA tester prior to interfacing with STE/ICE. Nonconformance to 3.25.12 shall constitute failure of this test.

4.4.8 DCA-STE/ICE interface. Upon completion of inspection and tests specified in 4.4.7, test the DCA for all measurements specified in 3.25.12 using the STE/ICE. Nonconformance to 3.25.12 or failure of any component or wiring in the DCA harness which results in inability to obtain the measurements previously recorded with the STE/ICE shall constitute failure of this test.

4.4.9 Towing lugs. The roller towing lugs shall be tested for compliance to 3.28.

4.4.10 Engine cooling. The contractor shall certify that the engine cooling meets the requirements of 3.12.3.2. Failure to provide certification will be cause for rejection.

4.4.11 Pad foot shell. The pad foot shell kit installation shall be tested for compliance to 3.23. Failure to pass any phase of this test shall be cause for rejection.

4.5 Inspection of packaging. The preservation, packing, and marking shall be examined in accordance with the quality assurance provisions of MIL-R-3075.

5. PACKAGING

5.1 Preservation, packing and marking. Each complete roller shall be preserved, packed, and marked in accordance with MIL-R-3075. The level of preservation shall be level A, B, or C and the level of packing shall be level A, B, or C as specified (see 6.2)

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The roller is intended for use in compacting granular and cohesive soils.

6.2 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article inspection is not required (see 3.2).
d. Time frame for submission of first article roller and number of rollers required. (see 3.2)
e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. When the Government will furnish the property specified in 3.33.
g. Color when other than as specified (see 3.35).
h. When roller will be overcoated with camouflage pattern (see 3.35).
i. When camouflage patterns will be provided (see 3.35.2 and 6.2.1).
j. Level of preservation and packing required (see 5.1).

6.2.1 Consideration of data requirements. The following data requirements must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
a. 3.35.2	DI-MISC-80192	Color photograph prints
b. 3.35.2	DI-MISC-80176	Camouflage line art data

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.33.

6.5 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical

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manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.6 Noise levels. When the required maximum noise limit(s) are established to be technically infeasible and beyond the state-of-the-art for the item/system being procured, the contracting officer may request additional information on noise sources. As such, documentation shall be furnished to the procuring activity for consideration of whether or not increase of the required limit(s) is justified and how noise levels may best be minimized. Documentation shall contain technically defensible data including technically supported design considerations, technically supported design recommendations for noise reduction, and technically supported predictions of the resultant noise levels. Clearly stated, convincing, and technically/fiscally supported trade-off analyses of noise control benefit against other design requirements such as weight, access, etc., shall be provided.

6.7 Subject term (key word) listing.

Pad foot
Segmented pads
Strike-off blade

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

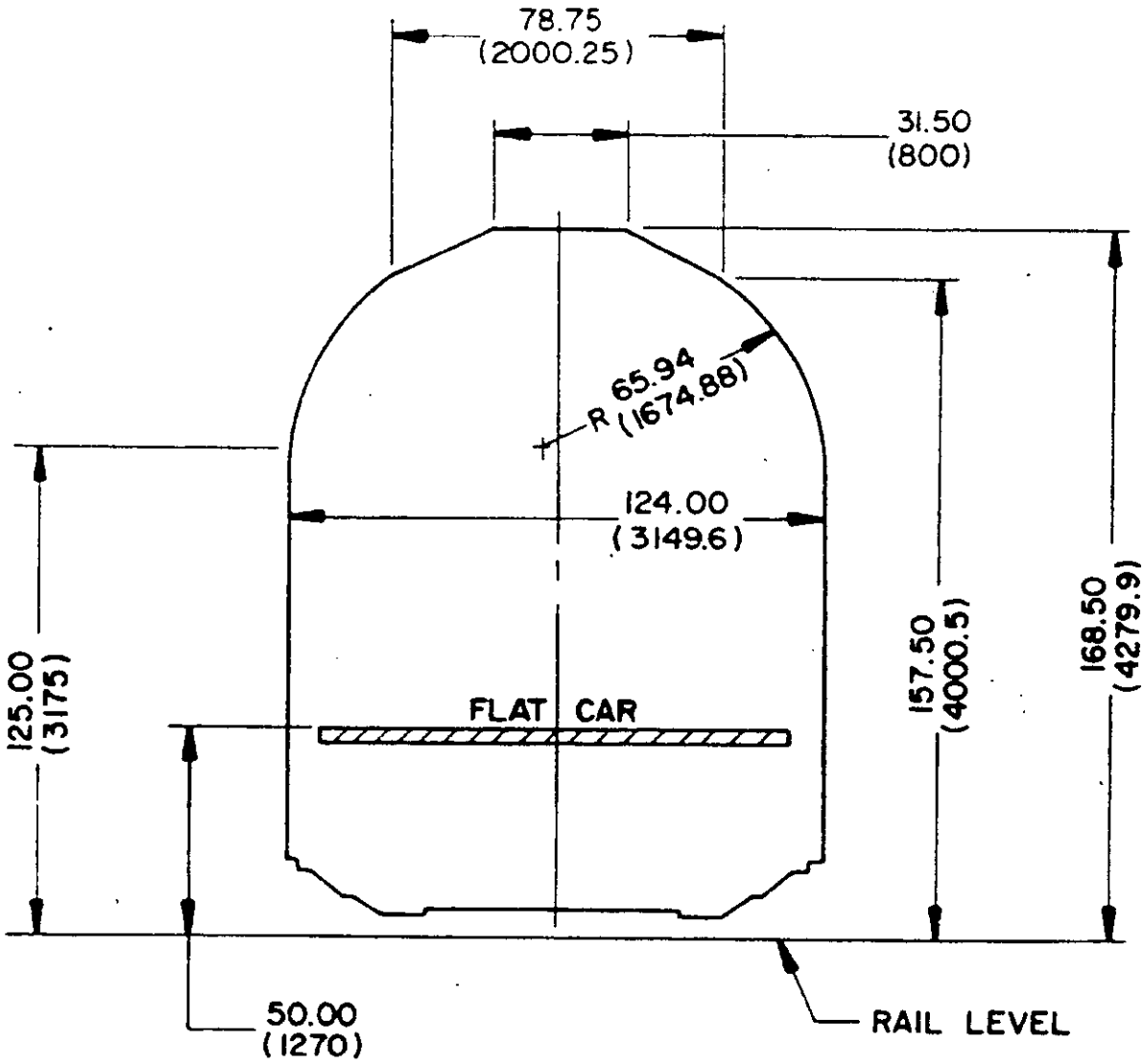
Custodian:
Army - ME

User activity:
Army - AT, MT

Preparing activity:
Army - ME

Project 3895-A320

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NOTE: DIMENSIONS SHOWN: INCHES (MILLIMETERS)

FIGURE 1. Gabarit International De Chargement (GIC) clearance diagram.

X - 4093 C

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-52934B(ME)	910211

3. DOCUMENT TITLE

Roller, Vibratory, Self-Propelled

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:	

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